Coursework Portfolio 1 — Week 9 Lab Report

Vulnerability Scanning & Mitigation Using Nessus

Student ID: SHE23601567

Module Code: CMP-X305-0

Submission Deadline: 31 March 2026

University of Roehampton London

Contents

Vulnerability Scanning & Mitigation Using Nessus 1
Introduction 5
Step 1: What is Nessus? 6
Step 2: Updating Kali Linux System Packages 7
Step 3: Start Nessus 8
Step 4: Scan two Virtual Machines 9
Kali Linux Scan Results 9
Ubuntu Server Scan Results 11
CVSS Scores 13
Conclusion 14
Reflection 15

Table of Figures

Figure 1: Nessus installer package confirmed in the Downloads directory. 6
Figure 2: System packages updated and upgraded before Nessus installation. 7
Figure 3: System upgrade completing successfully 7
Figure 4: Nessus 10.11.3 installed via dpkg with all integrity checks passing. 7
Figure 5: Nessus service started via systemctl. 8
Figure 6: Nessus web interface accessible at localhost:8834. 8
Figure 7: Nessus initialising and downloading plugins. 8
Figure 8: Nessus Essentials Plus dashboard ready to use. 8
Figure 9: Nessus scan configured targeting 192.168.123.100 9
Figure 10: Scan completed revealing 68 vulnerabilities across critical, high, medium, low and info severities. 9
Figure 11: Nessus report summary showing vulnerability severity distribution for 192.168.123.100. 10
Figure 12: Ubuntu Server VM running in VirtualBox. 11
Figure 13: Figure 13: Ubuntu VM IP address identified as 192.168.123.30. 11
Figure 14: Ping confirming connectivity to the WordPress VM at 192.168.123.30. 11
Figure 15: Nessus Ubuntu scan completed revealing 21 vulnerabilities in 8 minutes 12
Figure 16: Nessus report showing 1 medium and 27 informational findings for the WordPress VM. 12
Figure 17: CVSS v3.0 base score of 7.8 indicating a high severity vulnerability. 13
Figure 18: CVSS v3.0 base score of 5.9 indicating a medium severity vulnerability. 13

Introduction

In this lab, I installed and configured Nessus on Kali Linux to perform vulnerability scans on two virtual machines. The goal was to identify security weaknesses, understand their severity using CVSS scores, and document the findings. This gave me practical experience using an industry standard vulnerability scanner in a controlled lab environment.

Step 1: What is Nessus?

Nessus is a vulnerability assessment tool developed by Tenable that is widely used in cybersecurity to identify security weaknesses in computer systems and networks. It is one of the most popular vulnerability scanners used by security professionals and organisations worldwide.

Nessus works by scanning systems and comparing the results with a large database of known vulnerabilities. These vulnerabilities are identified using the Common Vulnerabilities and Exposures (CVE) database. The scanner checks for security issues such as missing patches, misconfigurations, weak passwords, and insecure network services.

Nessus can perform several types of security checks including:

- Identifying local and remote vulnerabilities
- Performing configuration and compliance audits
- Detecting default credentials
- Scanning web applications for security issues

By identifying vulnerabilities before attackers can exploit them, Nessus helps organisations strengthen their overall security posture.

Figure 1: Nessus installer package confirmed in the Downloads directory.

Step 2: Updating Kali Linux System Packages

Before installing Nessus, the Kali Linux system was updated to ensure that all packages and dependencies were up to date. Updating the operating system helps prevent compatibility issues and ensures the latest security patches are applied.

The following command was executed in the terminal:

sudo apt update && sudo apt upgrade

The "*apt update*" command retrieves the latest package lists from the Kali repositories, while "*apt upgrade*" installs the newest versions of installed packages. Keeping the system updated is an important security practice before installing new software tools.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update && sudo apt upgrade -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1768 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer require
d:
  curlftpfs                libmplex2-2.1-0t64  mesa-vdpau-drivers
  libaudio2                libmupdf25.1        pocketsphinx-en-us
  libavfilter10            libplacebo351       python3-aiocache
  libavformat61            libpocketsphinx3    python3-aiomcache
  libconfig-inifiles-perl  libpostproc58       python3-fs
  libcrypt-dev             librubberband2      python3-terminaltables
  libdlt2                  libsframe2          python3-wapiti-arsenic
  libfuse2t64              libsnmp40t64        python3-yaswfp
  libgav1-1                libsphinxbase3t64   ruby-unf-ext
  libgdk-pixbuf2.0-bin     libswscale8         vdpau-driver-all
  libmjpegutils-2.1-0t64   libtsk19t64
  libmpeg2encpp-2.1-0t64   libvdpau-va-gl1
Use 'sudo apt autoremove' to remove them.

Upgrading:
```

Figure 2: System packages updated and upgraded before Nessus installation.

```
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
        This may take some time... done.
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.5+dfsg-4)…
Processing triggers for initramfs-tools (0.150)…
update-initramfs: Generating /boot/initrd.img-6.18.12+kali-amd64
```

Figure 3: System upgrade completing successfully

After downloading the Nessus package, the terminal was used to navigate to the Downloads folder. The package was installed using the command:

"sudo dpkg -i Nessus-10.11.3-ubuntu1604_amd64.deb"

This installed Nessus on the Kali Linux system and prepared it to run as a background service.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo dpkg -i Nessus-10.11.3-ubuntu1604_amd64.deb

Selecting previously unselected package nessus.
(Reading database… 438669 files and directories currently installed.)
Preparing to unpack Nessus-10.11.3-ubuntu1604_amd64.deb…
Unpacking nessus (10.11.3)…
Setting up nessus (10.11.3)…
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
```

Figure 4: Nessus 10.11.3 installed via dpkg with all integrity checks passing.

Step 3: Start Nessus

Once the Nessus service was running, the web interface was accessed using a browser inside Kali

Linux by navigating to:

https://localhost:8834/

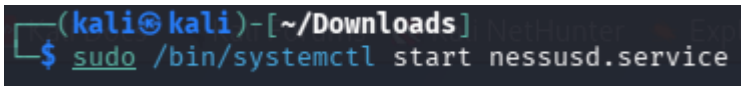

Figure 5: Nessus service started via systemctl.

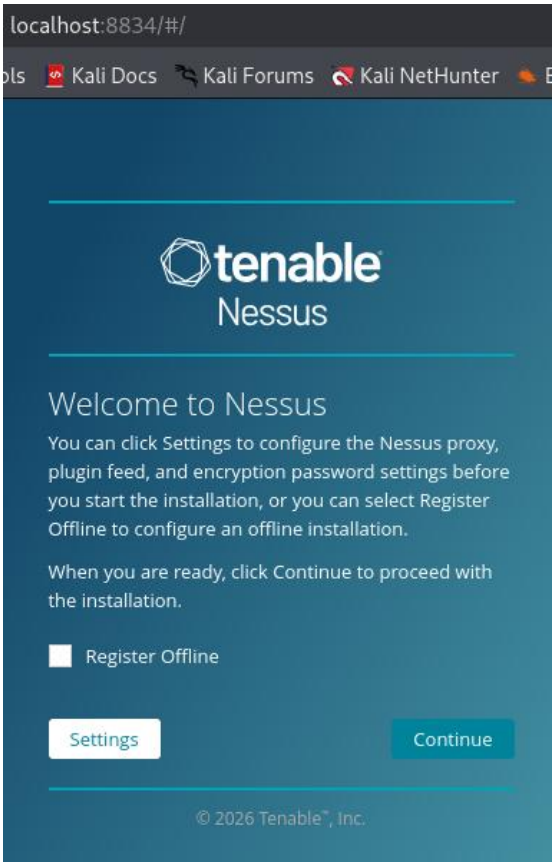

Figure 6: Nessus web interface accessible at localhost:8834.

After accessing the Nessus web interface, Nessus Essentials was selected and activated using the license code obtained from the Tenable Community account. During activation, Nessus downloaded the latest vulnerability plugins to the local system.

This process enables the scanner to perform accurate vulnerability assessments on target systems. The setup also requires creating an administrator account to manage scans and reports.

Figure 7: Nessus initialising and downloading plugins.

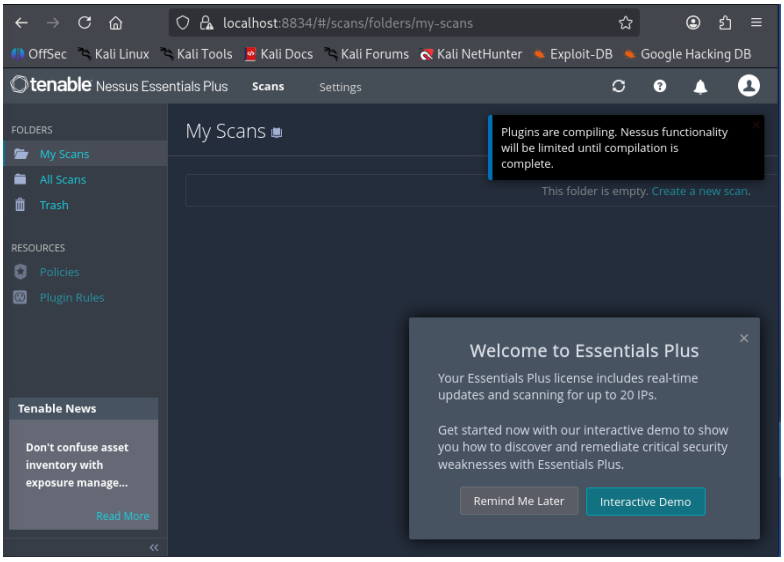

Figure 8: Nessus Essentials Plus dashboard ready to use.

Step 4: Scan two Virtual Machines

Two virtual machines were scanned using the Nessus vulnerability scanner.

The first scan targeted the Kali Linux virtual machine, which is the system where Nessus was installed. The second scan targeted an Ubuntu Server virtual machine running a WordPress environment.

The purpose of these scans was to identify possible security weaknesses and understand the security posture of the systems

Kali Linux Scan Results

The Kali Linux system was scanned using Nessus with the target IP address 192.168.123.100.

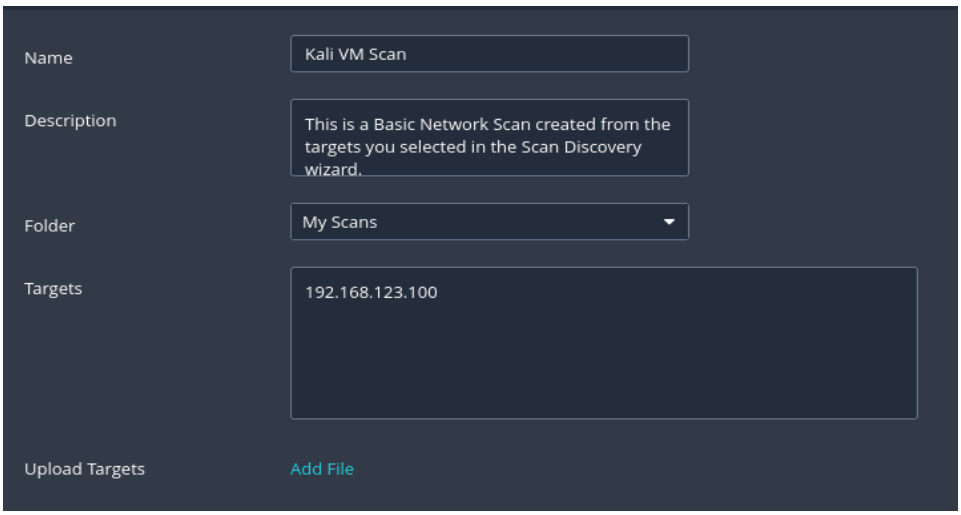

Figure 9: Nessus scan configured targeting 192.168.123.100

The scan results identified several vulnerabilities of different severity levels

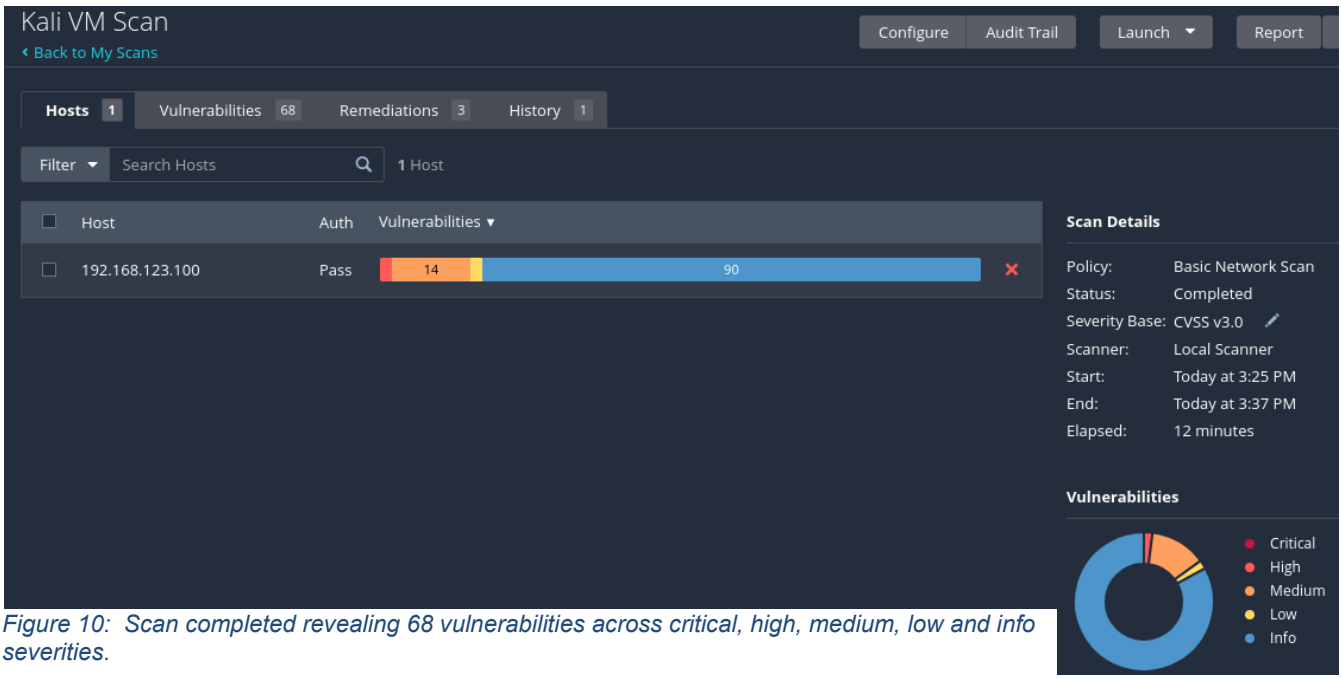

Figure 10: Scan completed revealing 68 vulnerabilities across critical, high, medium, low and info severities.

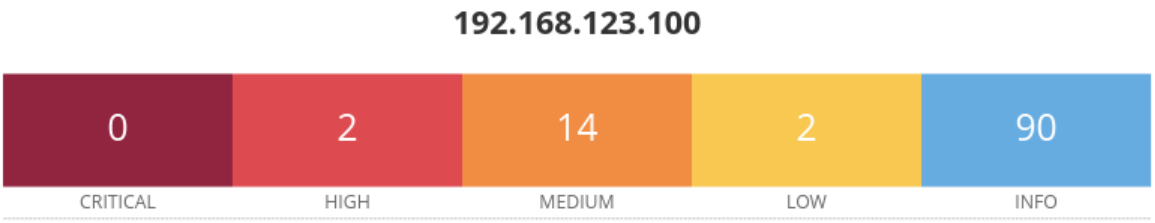

Figure 11: Nessus report summary showing vulnerability severity distribution for 192.168.123.100.

The scan results show that the Kali Linux system contains no critical vulnerabilities, which means there are no extremely dangerous issues that require immediate action.

However, the scan detected 2 high-severity vulnerabilities and 14 medium-severity vulnerabilities. These types of vulnerabilities could potentially be exploited by attackers if they are not addressed.

Many of the findings were informational vulnerabilities (90 findings). Informational findings are not direct security threats. Instead, they provide details about the system such as:

- Operating system version
- Open network services
- Running software version

This information helps security professionals understand how a system is configured and whether any security improvements are needed.

Overall, the Kali system shows moderate security risks, mainly due to the medium-level vulnerabilities identified.

Ubuntu Server Scan Results

The scan targeted the Ubuntu server virtual machine, which had the IP address 192.168.123.30.

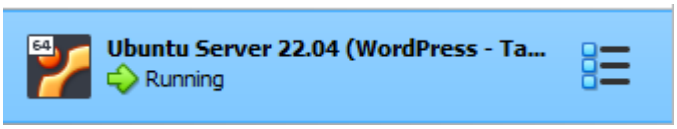

Figure 12: Ubuntu Server VM running in VirtualBox.

```
student@wordpress:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
o
    link/ether 08:00:27:84:10:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.123.30/24 brd 192.168.123.255 scope global
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe84:10bc/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
o
    link/ether 08:00:27:d4:4c:5c brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic e
       valid_lft 86398sec preferred_lft 86398sec
    inet6 fe80::a00:27ff:fed4:4c5c/64 scope link
       valid_lft forever preferred_lft forever
student@wordpress:~$
```

Figure 13: Figure 13: Ubuntu VM IP address identified as 192.168.123.30.

Before starting the vulnerability scans, a ping test was performed from the Kali Linux machine to the Ubuntu virtual machine. This step is to ensure that the two virtual machines can communicate over the network in order to scan.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.123.30
PING 192.168.123.30 (192.168.123.30) 56(84) bytes of data.
64 bytes from 192.168.123.30: icmp_seq=1 ttl=64 time=2.89 ms
64 bytes from 192.168.123.30: icmp_seq=2 ttl=64 time=0.412 ms
64 bytes from 192.168.123.30: icmp_seq=3 ttl=64 time=0.483 ms
```

Figure 14: Ping confirming connectivity to the WordPress VM at 192.168.123.30.

The scan results show that the Ubuntu server has no critical or high-severity vulnerabilities, which indicates that the system is relatively secure.

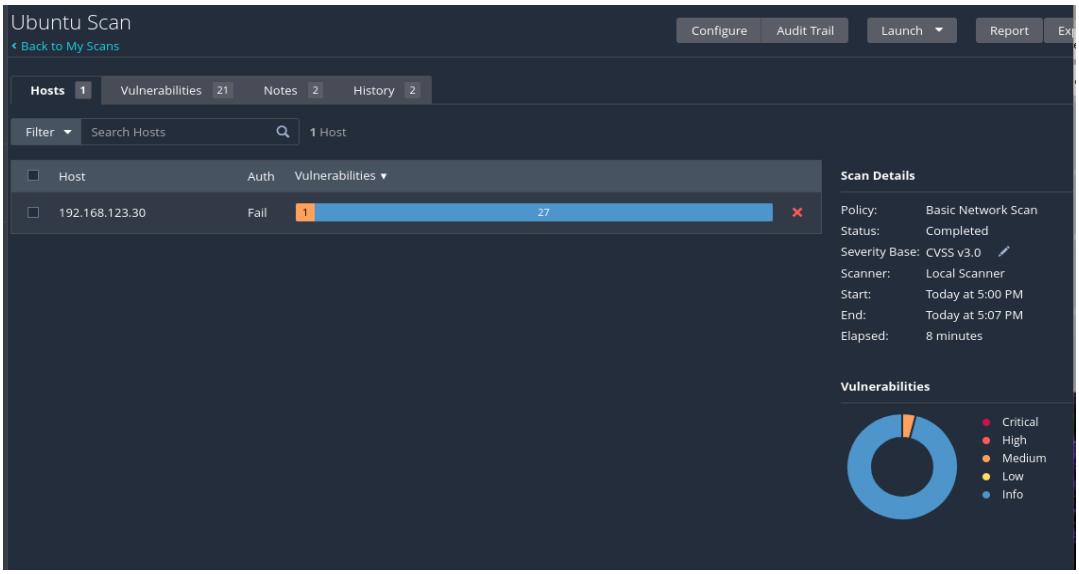

Figure 15: Nessus Ubuntu scan completed revealing 21 vulnerabilities in 8 minutes

Only one medium-severity vulnerability was detected. Medium vulnerabilities represent moderate security risks that should be reviewed and patched but are not considered immediately dangerous

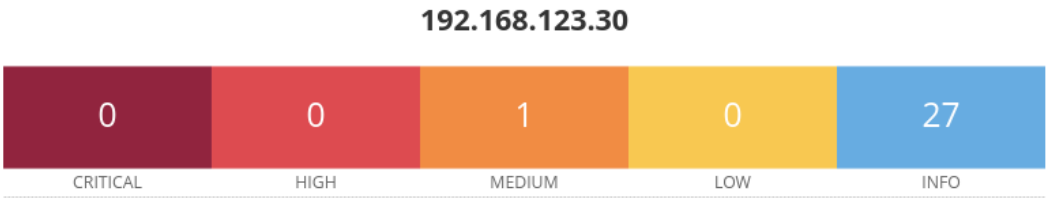

Figure 16: Nessus report showing 1 medium and 27 informational findings for the WordPress VM.

Most findings were informational vulnerabilities (27 findings). These simply provide information about the system such as:

- Operating system identification
- Running services
- Network configuration

Overall, the Ubuntu server shows low security risk, as only one medium-severity vulnerability was detected.

CVSS Scores

Nessus uses the CVSS (Common Vulnerability Scoring System) to measure the severity of vulnerabilities.

CVSS provides a score between 0 and 10, which helps security teams understand how serious a vulnerability is.

CVSS Score	Severity	Meaning
0.0	None	No security risk
0.1 - 3.9	Low	Small security issue
4.0 - 6.9	Medium	Moderate security risk
7.0 - 8.9	High	Serious vulnerability
9.0 - 10	Critical	Extremely dangerous vulnerability

Kali had a CVSS base score of 7.8 which indicates a high-severity vulnerability, meaning the issue could potentially be exploited by an attacker and may have significant impact on the system's security. Vulnerabilities with this score should be addressed as a priority because they could allow unauthorized access, data exposure, or disruption of system services if left.

CVSS v3.0 Base Score

7.8 (CVSS:3.0/AV:L/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H)

Figure 17: CVSS v3.0 base score of 7.8 indicating a high severity vulnerability.

The vulnerability found on the Ubuntu server had a CVSS score of 5.9 which falls into the medium severity category. This means the issue should be addressed but does not pose an immediate critical threat.

CVSS v3.0 Base Score

5.9 (CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:H/A:N)

Figure 18: CVSS v3.0 base score of 5.9 indicating a medium severity vulnerability.

Conclusion

The Nessus scans successfully identified vulnerabilities on two virtual machines within the lab environment.

The Kali Linux system showed a large number of vulnerabilities, including two high-severity issues and several medium-severity findings. In contrast, the Ubuntu server system had only one medium-severity vulnerability and several informational findings.

This demonstrates how vulnerability scanners such as Nessus can help security professionals identify weaknesses, prioritise risks, and improve the security posture of systems.

Reflection

This lab gave me hands-on experience using Nessus to scan real systems and interpret the results. Setting it up from the command line and accessing it through the browser made the process feel professional and realistic.

Comparing the two scan results was the most valuable part. The Kali Linux system had more vulnerabilities than expected, including 2 high severity findings, which showed me that even a security focused operating system can have weaknesses. The Ubuntu server was much cleaner with only one medium severity vulnerability, demonstrating that regular maintenance keeps risk low.

Understanding CVSS scores was a key takeaway. Having a numerical score gives security teams a clear way to prioritize what needs fixing first rather than treating all vulnerabilities the same.

Overall Nessus was an effective and straightforward tool. The visual reports made findings easy to understand and present. This lab improved my ability to assess system security in a structured and professional way.